Operating Systems Midterm Exam
Spring 2025

Name (First and Last): ____

	Points (15)
Multiple Choice	

Question	Topic	Points (0-5)
1	OS Kernel	
2	Process Model	
3	Scheduler	
4	Virtual Addresses	
5	Page Replacement Algorithms	

	Points (out of 40)
Total	

1.. Which component of the Xinu operating system tracks process metadata?

- ☐ A) The process table
- ☐ B) The readylist
- ☐ C) The proc file
- ☐ D) The interrupt vector

2. What is the fundamental role of an operating system?

- ☐ A) To allow direct hardware access by applications
- ☐ B) To compile and run user programs
- ☐ C) To manage hardware resources and provide services to applications
- ☐ D) To perform data backup operations automatically

3. Which of the following best describes a lightweight process?

- ☐ A) A process that consumes minimal CPU time
- ☐ B) A thread that shares its address space with other threads
- ☐ C) A process that runs in the background with low priority
- ☐ D) A process that runs without needing memory allocation

4. In the Producer-Consumer problem, what happens if the producer tries to add to a full buffer?

- ☐ A) The producer overwrites existing data
- ☐ B) The producer goes to sleep until there is space
- ☐ C) The producer starts consuming items instead
- ☐ D) The producer terminates

5. A scheduling algorithm is being tested for an interactive system where users frequently input commands that should be processed quickly. The system administrator finds that some short tasks are experiencing excessive waiting times due to longer-running background processes. What scheduling algorithm would best address this issue, and why?

- ☐ A) Round Robin, because it enforces time-sharing by giving each process a fair slice of CPU time
- ☐ B) First-Come, First-Served (FCFS), because it ensures fairness by executing jobs in the order they arrive
- ☐ C) Shortest Job Next (SJN), because it prioritizes jobs that require the least CPU time
- ☐ D) Priority Scheduling, because it assigns lower priority to short tasks

6. Which of these is a primary cause of internal fragmentation?

- ☐ A) Fixed-size memory allocation
- ☐ B) Variable-size memory allocation
- ☐ C) Paging
- ☐ D) Memory swapping

7. What type of memory is used by the CPU for immediate processing and is the fastest?

- ☐ A) Flash
- ☐ B) Cache
- ☐ C) RAM
- ☐ D) SSD

8. What is an example of spatial multiplexing in an operating system?

- ☐ A) A process using a shared memory segment for communication
- ☐ B) Many virtual pages share a small number of physical page frames
- ☐ C) The CPU switching between different processes at rapid intervals
- ☐ D) A single process running continuously without interruption

9. What information is saved during a context switch?

- ☐ A) The process's CPU registers and program counter
- ☐ B) The entire contents of RAM
- ☐ C) Only the process ID
- ☐ D) The process's priority level

10. In a modern operating system, processes are typically isolated from each other using virtual memory. However, some system designs allow processes to share memory for efficiency. What mechanism allows two or more processes to share a section of their address space while maintaining separate execution states?

- ☐ A) Interprocess Communication (IPC) using pipes
- ☐ B) Shared Memory Segments managed by the operating system
- ☐ C) Copy-on-write memory allocation
- ☐ D) Process forking with independent memory spaces

11. Which of the following is an example of a race condition?

- ☐ A) Two processes executing without sharing resources
- ☐ B) Two threads accessing a shared variable without synchronization
- ☐ C) A single-threaded program running on a single CPU
- ☐ D) A process waiting for another process to terminate

12. What happens when a program accesses a page that is not in memory?

- ☐ A) The OS kills the process immediately
- ☐ B) A segmentation fault occurs
- ☐ C) A page fault is triggered
- ☐ D) The program continues executing normally

13. A process control block (PCB) is a data structure used by the operating system to store information about a process. Which of the following is not typically stored in a PCB?

- ☐ A) Process ID and scheduling information
- ☐ B) CPU register values and program counter
- ☐ C) System-wide file permission table
- ☐ D) Memory management information

14. What is the primary function of Direct Memory Access (DMA) in an operating system?

- ☐ A) To manage virtual memory allocation
- ☐ B) To transfer data between peripherals and memory without CPU intervention
- ☐ C) To optimize file system performance
- ☐ D) To allow applications to access hardware directly

15. How does the Xinu readylist structure differ from a traditional scheduling queue?

- ☐ A) It is a FIFO queue
- ☐ B) It is a priority queue that dynamically reorders based on process priority
- ☐ C) It stores terminated processes
- ☐ D) It holds only I/O-bound processes

1. OS Kernel

System calls allow user applications to request privileged operations from the OS kernel. Describe the steps involved in handling a system call from a user process. Consider user application reading or writing to a file as an example system call, for example.

2. Process Model

What is the purpose of each of the following fields in the Process Table Entry structure in Xinu?

1. prstate
2. prprio
3. prstkptr
4. prstkbase
5. prstklen
6. prhasmsg
7. prmsg

3. Scheduler

You have been tasked with implementing a dynamic quantum in Xinu's scheduling algorithm. Higher priority processes will be given more CPU time to run before being interrupted.

a) Explain how the current Xinu scheduler can experience a convoy effect.
b) What impact will this change have on the number of context switches and throughput?

4. Virtual Addresses

The Translation Lookaside Buffer (TLB) was added into the CPU's memory management unit by CPU manufacturers.

a) What is stored in the TLB?
b) Why was the TLB added at the hardware level?

5. Page Replacement Algorithm

A system uses a Least Recently Used (LRU) page replacement algorithm and has 4 available page frames. Consider the following sequence of page references for (b):

1, 3, 0, 3, 5, 6, 3, 0, 1, 4, 3, 0, 6

(a) Explain how the Least Recently Used (LRU) algorithm works.
(b) Simulate the page replacement process and determine the total number of page faults from the sequence above.